

PAVELETS, Russian Federation

WMO: 278230

Lat: 53.7840N

Lon: 39.2379E

Elev: 209

StdP: 98.84

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-20.8	-26.8	0.3	-23.5	-23.8	0.5	-20.6	10.4	-3.9	9.5	-3.0	1.5	0	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.9	19.3	28.3	18.4	26.6	17.9	20.9	27.3	19.8	25.7	18.9	24.3	2.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	23.6	17.7	13.0	22.4	16.7	12.2	21.5	61.2	27.6	57.6	25.8	54.5	24.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.8	7.6	6.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-27.0	22.5	2.0	1.9	-28.4	23.8	-29.6	24.9	-30.8	26.0	-32.2	27.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-8.1	-6.2	-2.5	6.5	13.2	17.3	19.9	17.0	11.5	4.6	-1.7	-5.7
	DBStd	10.92	6.62	5.73	4.93	4.75	4.82	4.18	3.23	3.63	4.05	4.66	5.66	5.94
	HDD10.0	2592	562	453	386	123	24	1	0	0	31	174	351	487
	HDD18.3	4809	820	686	644	353	168	71	20	69	207	424	601	746
	CDD10.0	971	0	0	0	19	124	219	307	217	75	8	0	0
	CDD18.3	147	0	0	0	0	9	39	69	28	2	0	0	0
	CDH23.3	1304	0	0	0	0	91	351	602	250	10	0	0	0
	CDH26.7	339	0	0	0	0	14	89	179	57	0	0	0	0
(14) Wind	WSAvg	2.4	3.2	3.3	3.0	2.4	2.3	1.8	1.6	1.5	1.7	2.4	2.9	3.1
(15) Precipitation	PrecAvg	542	35	27	28	37	45	59	76	58	51	44	45	43
	PrecMax	731	91	55	54	86	95	129	162	108	113	87	92	85
	PrecMin	371	6	1	8	6	6	8	4	8	4	10	15	9
	PrecStd	95	21	16	13	18	23	32	40	30	29	22	21	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	3.3	11.2	22.0	28.2	31.3	32.2	30.6	25.9	19.0	11.2	6.0
		MCWB	1.1	2.2	7.3	11.9	16.2	18.8	20.0	19.4	15.4	12.2	9.5	5.5
	2%	DB	1.4	2.0	7.5	19.1	25.8	29.1	30.3	28.3	22.9	15.3	8.7	3.2
		MCWB	0.7	1.2	4.3	10.9	15.6	18.1	19.6	18.6	15.5	12.0	7.6	2.5
	5%	DB	0.8	1.2	5.4	17.0	23.4	27.5	28.8	26.1	20.4	13.2	6.7	1.6
		MCWB	0.2	0.5	3.0	9.8	14.1	17.9	19.3	17.8	14.0	10.7	5.8	1.0
	10%	DB	0.2	0.6	3.5	14.6	21.7	25.3	26.8	24.1	18.3	11.4	5.0	0.7
		MCWB	-0.3	0.0	1.7	8.8	13.6	17.3	18.7	17.2	13.2	9.4	4.1	0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.5	7.7	13.4	17.7	22.1	22.9	21.2	17.5	14.5	9.8	5.5
		MCDB	1.8	2.9	10.7	18.4	24.3	28.3	28.9	27.5	23.6	16.3	10.9	5.8
	2%	WB	0.9	1.4	5.0	11.8	16.5	20.2	21.4	19.7	16.2	12.5	7.7	2.7
		MCDB	1.2	1.7	6.8	16.8	23.1	26.3	27.9	26.0	20.7	14.0	8.5	3.0
	5%	WB	0.4	0.7	3.2	10.6	15.6	19.0	20.2	18.6	14.9	11.2	5.9	1.2
		MCDB	0.8	1.0	5.0	16.0	22.0	25.2	26.3	24.2	18.7	13.3	6.6	1.5
	10%	WB	-0.3	0.2	1.8	9.3	14.5	17.8	19.5	17.7	13.9	9.5	4.1	0.4
		MCDB	0.2	0.6	3.1	14.2	20.2	23.7	24.9	22.9	17.3	11.3	4.8	0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	5.9	6.7	9.1	10.5	10.1	10.0	10.1	8.7	6.7	4.6	4.8
		MCDBR	3.8	4.4	8.7	13.0	13.0	12.7	12.9	13.1	11.8	9.4	6.2	4.0
		MCWBR	3.7	4.0	6.0	6.8	5.6	5.0	4.4	5.3	5.6	6.2	5.5	3.7
	5% WB	MCDBR	4.0	4.0	7.9	11.5	11.2	11.6	10.4	11.4	9.8	8.2	6.1	3.7
		MCWBR	4.1	3.9	5.7	6.6	5.5	5.4	4.8	5.5	5.5	6.2	5.6	3.6
(40) Clear-Sky Solar Irradiance	taub		0.290	0.331	0.352	0.424	0.390	0.382	0.416	0.435	0.382	0.356	0.326	0.294
	taud		2.115	2.035	2.087	2.096	2.311	2.350	2.272	2.205	2.334	2.336	2.235	2.133
	Ebn at Noon		643	729	811	798	857	867	826	782	783	714	601	553
	Edh at Noon		69	104	123	141	120	117	124	125	97	78	62	56
(44) All-Sky Solar Radiation	RadAvg		0.84	1.84	3.38	4.20	5.62	5.78	5.68	4.71	3.05	1.50	0.64	0.46
	RadStd		0.16	0.19	0.42	0.36	0.47	0.62	0.53	0.48	0.38	0.23	0.12	0.13

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	-133	-209	+132	N/A

Nomenclature: See separate page